

Comprehensive Project Report: Customer Churn Classification

1. Project Title & Overview

This project focuses on building a predictive machine learning model to identify customers at high risk of churning from a telecom service. By accurately identifying these individuals, the business can take proactive retention measures and minimize revenue loss.

2. Dataset Description

- **Source:** `customer_churn_data.csv`
- **Features:** Includes customer demographics, account information, and subscribed services (e.g., tenure, MonthlyCharges, Contract type).
- **Target Variable:** `Churn` (Binary: Yes/No indicating if the customer left).

3. Setup & Installation

Prerequisites

- Python 3.8+

Installation

Navigate to this directory and install the required dependencies:

```
pip install -r requirements.txt
```

4. Methodology & Pipeline

- **Data Preprocessing:**
 - Dropped non-predictive columns (`customerID`).
 - Handled missing values (imputed median for `TotalCharges`).
 - Scaled numerical features using `StandardScaler`.
 - Encoded categorical variables using `OneHotEncoder`.
- **Model Architecture:** We trained and compared three different models to find the best fit:
 1. Logistic Regression (Baseline)
 2. Random Forest
 3. Gradient Boosting Classifier

5. Results & Evaluation

The models were evaluated using 5-fold stratified cross-validation and a hold-out test set.

- **Metrics Evaluated:** Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

- **Winning Model:** The **Random Forest** model was selected as the best-performing model based on achieving the highest ROC-AUC score. (Refer to *models_performance_comparison.png* and *Screenshot.png* in this folder for visual performance comparisons).

6. Usage / How to Run

You can run the full pipeline either through the Jupyter Notebook or the provided Python script:

```
# To run the script version:  
python task1.py
```

7. Appendix: Screenshots & Visualizations

(Attach *models_performance_comparison.png* in the space below)

[Space for models_performance_comparison.png]

(Attach *Screenshot.png* in the space below)

